

Project Title: Earthquake Vulnerable Building Detection & Area-Based Risk Analysis System for Bangladesh

Administrative System

● Building Data Management

- Add / Edit / Delete Building Records
- Approve Newly Submitted Building Information
- Store Approved Building Plans

● Seismic Zone & BNBC Data Management

- Add / Update Earthquake Zone Classification
- Manage BNBC (Bangladesh National Building Code) Update Years
- Apply Risk Multipliers Based on Zone

● Risk Parameter Configuration

- Set Weight for Construction Year
- Set Weight for Floor Count
- Set Weight for Soil Type
- Set Weight for Building Materials
- Set BNBC Compliance Value

● User Account Management

- Approve Structural Engineers & Inspectors
- Manage Citizens & Emergency Responders
- Role-Based Access Control

● Support & Communication

- Admin Support Chat
 - Emergency Alert Broadcasting
 - Send Area Risk Notifications
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User Account System

- **Sign Up**

- Citizen
- Structural Engineer
- Inspector
- Emergency Responder

- **Login**

- Secure Role-Based Dashboard

- **Profile Management**

- Update Personal Information
- Upload Verification Documents (Engineers/Inspectors)

- **Building Submission History**

- View Submitted Buildings
- View Risk Reports

- **Area Risk View**

- Check Risk Score of Own Area
- Download Risk Summary Report

Building Risk Analysis System

- **Submit Building for Risk Analysis**

- Enter Building Details
 - Construction Year
 - Number of Floors
 - Building Materials
 - Soil Type
 - BNBC Compliance
- Upload Structural Images
- Mark Location on Map

- **Structural Weakness Scoring Algorithm**

- Calculate Building Risk Score
- Apply Seismic Zone Multiplier
- Categorize as:
 - Safe
 - Moderate Risk
 - High Risk

- **Old vs Updated BNBC Comparison**

- Check Construction Year
- Compare with Major BNBC Update Years
- Increase Risk Weight if Non-Compliant

- **Illegal Floor & Structural Modification Detection**

- Compare Approved Plan vs Actual Floors
- Detect Unauthorized Extensions
- Automatically Increase Risk Score

- **Inspection History Management**

- Upload Damage Photos (Cracks, Column Failure, Foundation Damage)
- Maintain Multiple Inspection Records
- Track Structural Condition Over Time

Area-Based Risk Analysis

- **Risk Heatmap & Area Filtering**

- Interactive Bangladesh Map
- Color-Coded Risk Zones
 - Green (Safe)
 - Yellow (Moderate Risk)
 - Red (High Risk)
- Filter by City / Area / Ward
- Filter by Risk Score Range

- **Smart Priority Ranking**

- Rank Top Most Dangerous Buildings
- Identify High-Risk Clusters
- Suggest Priority Inspection Order

- **Emergency Preparedness Score**

- Check Availability of Fire Exits
- Check Emergency Stairs
- Check Open Assembly Areas
- Evaluate Rescue Accessibility

- **Area Living Suitability Index**

- Combine:
 - Average Building Risk Score
 - Emergency Preparedness Score
 - Seismic Zone Classification
 - Classify Area as:
 - Suitable for Living
 - Moderately Safe
 - Highly Vulnerable
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Emergency Response System

- **Assign Emergency Responders**

- View High-Risk Buildings
- Assign Rescue Teams
- Monitor Emergency Status

- **Real-Time Alert System**

- Send Earthquake Warnings
 - Notify High-Risk Area Residents
 - Provide Evacuation Guidelines
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Expected System Outcome

- Identify Earthquake Vulnerable Buildings in Bangladesh
- Provide Scientific Risk Scoring
- Help Government & Authorities Take Preventive Action
- Increase Public Awareness
- Improve Urban Safety Planning
- Support Disaster Management Decision Making
